

Education

Year	Degree/Examination	Board/University	%/ CGPA
2021-2025	Bachelor of Technology(ECE)	Indian Institute of Technology Bhubaneswar	7.10/10
2021	Grade 12	Narayana Junior College	97.9%
2019	Grade 10	Narayana High School	10/10

Experience

Research Intern(May 2024 - july 2024)

Under the guidance of Dr Nitya Tiwari IIT Bhubaneswar

MCNET: Multichannel Speech Enhancement Using Multi-Cue Fusion[PyTorch]

- Developed a multi-cue fusion network (MCNET) for real-time multichannel speech enhancement, combining spectral and spatial information using LSTM-based networks for temporal dependency modeling.
- Achieved significant performance improvements, with 3.38 NB-PESQ, 2.73 WB-PESQ, 97.6% STOI, and 19.6 SDR, outperforming existing methods like FT-JNF and Oracle MVDR in multiple metrics.

Projects

Image-to-Speech GenAI Tool Using LLM[GEN AI and Full Stack]

- Designed and deployed an end-to-end AI application that transforms images into narrated short stories, combining image captioning, creative storytelling (GPT-3.5), and text-to-speech synthesis, using LangChain, Hugging Face, and OpenAI APIs.
- Built a responsive, user-friendly web interface using Streamlit, enabling real-time user interaction and hosted it on Streamlit Cloud and Hugging Face Spaces.
- Optimized the end-to-end inference pipeline, reducing latency by 20%, enhancing real-time performance and user experience.

Video Evaluation Framework using Multiagent System and Computer Vision[Computer Vision]

- Designed and implemented a multiagent architecture for automated video evaluation, incorporating perception and reporting agents communicating through shared memory.
- Developed robust video analysis pipelines using OpenCV for temporal coherence, dynamic scene handling, and generalization assessment across diverse datasets.
- Integrated pretrained vision-language models (CLIP) to measure semantic consistency, achieving a scalable and modular evaluation system with reproducible Docker environments.

End-to-End Data Analytics on Ola Ride Data Using SQL and Power BI[SQL and PowerBI]

- Developed an end-to-end data analytics project using Ola's ride data, encompassing SQL querying, data cleaning, and Power BI dashboard creation.
- Analyzed over 100,000 daily entries across 19 columns, focusing on successful bookings, cancellations (by customer and driver), revenue by vehicle type, and customer behavior.
- Built interactive Power BI dashboards with real-time visualization of key metrics, including booking trends, cancellation rates, and driver performance, using advanced filtering and formatting techniques.

Technical Skills

- Programming Languages: Python,C,C++
- Other Technical Skills and Tools: SQL, Database Management Systems, Power BI,Tableau, Microsoft Excel, PowerPoint, AI(GenAI)Tools
- Framework/Libraries: Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, TensorFlow/keras, Pytorch

Positions of Responsibility

Core Head of Team Events and ManagementFeb 2022 – Mar 2024

Esummit, IIT Bhubaneswar

Nakshatra team memberNov 2021 – June 2022

Scholastic Achievements

- Secured a good rank in JEE ADVANCE.
- Secured 1734th Rank in A.S.Rao Talent Test and 497th in ISO.